

AZMAT HUSSAIN

AI / Machine Learning Engineer | Computer Vision & Sensor Fusion | Robotics & Embedded AI | Applied AI Solutions Architect & Product Lead

Karachi, Pakistan | +92 333 7069360 | hussainazmat025@gmail.com | linkedin.com/in/azmat-h | azmdriven.vercel.app | Google Scholar

PROFESSIONAL SUMMARY

Applied AI engineer, solutions architect, and product leader with 7+ years turning applied AI into systems that ship - including 3+ years leading cross-functional engineering, design, and research teams. Currently Product Architect at TPL Trakker, owning roadmap, architecture, and delivery of AI-driven sensing, computer-vision, and connected-hardware products. MS in Artificial Intelligence (NED University, 2024-2026); BE in Electrical & Electronics Engineering. Published researcher in navigation, sensor fusion, and applied deep learning (IEEE/ICIIE, MDPI x2). Strong across the full stack a modern AI product needs: Python and modern C/C++, TensorFlow / PyTorch / OpenCV, computer vision, multi-sensor fusion and state estimation, LLM / RAG / agentic workflows, and edge-AI deployment on Jetson, Raspberry Pi, and FPGA - equally at home defining the product, architecting the system, and writing the code that runs on the device. Open to opportunities globally; willing to relocate.

TECHNICAL SKILLS

Languages: Python, C / C++ (modern C++14/17), MATLAB, SQL, Bash

AI / ML: TensorFlow, PyTorch, scikit-learn, Hugging Face, OpenCV, NumPy / Pandas; CNNs, RNNs, transfer learning, model training, evaluation & benchmarking

Computer Vision & Perception: Object detection, image classification, pose estimation, deep-learning recognition pipelines, visual-inertial odometry, geometric CV, homography / calibration

Sensor Fusion & State Estimation: Kalman-family filters (EKF / UKF / error-state), Bayesian estimation, factor-graph optimisation, SLAM, multi-sensor calibration (camera / LiDAR / IMU / vehicle), ego-motion estimation, autonomous navigation

Generative AI: LLM APIs (OpenAI, Anthropic, Gemini) and open models, Retrieval-Augmented Generation, vector databases, embeddings, LangChain / LangGraph-style orchestration, multi-agent workflows, knowledge graphs (Neo4j), prompt engineering

Edge & Embedded AI: Jetson Xavier AGX / Orin / Nano, Raspberry Pi, FPGA, embedded Linux; TensorRT / TFLite / ONNX, quantisation, pruning, distillation; latency / memory / power / thermal profiling; real-time control, embedded firmware, signal processing

Hardware: PCB and electronics prototyping, digital fabrication, design for manufacture, mechanical-electrical integration, LoRa and low-power IoT device design

Backend, Infra & Robotics Tooling: FastAPI / Flask, REST APIs, Docker, Git, Linux, CI/CD, AWS / GCP (working knowledge); ROS / ROS2, CARLA, Gazebo, Eigen, Ceres / GTSAM-style optimisation, replay / bench / hardware-in-the-loop testing

Product & Leadership: Product strategy & roadmap ownership, PRDs, KPI definition, cross-functional team leadership, research-to-production translation, stakeholder & executive communication, technical due diligence

PROFESSIONAL EXPERIENCE

Product Architect (AI & Sensing Systems)

Apr 2024 - Present

TPL Trakker Ltd - Product Development & Innovation, Karachi

- Own end-to-end product strategy, architecture, and roadmap for AI-driven sensing, perception, and connected-hardware product lines, aligning engineering, design, and business stakeholders around a single technical vision.
- Architect perception and multi-sensor-fusion pipelines for a real-time connected-vehicle fleet, making hardware-software co-design calls that balance edge compute, power, connectivity, and latency budgets against AI model performance.
- Lead cross-functional teams across embedded engineering, computer vision, and product design to take applied-AI research from concept to field-deployed hardware-software systems.
- Delivered flagship products from concept to deployment - Audience Insight (on-device human detection and pose estimation for real-time analytics), Indoor Asset Tracking (sensor-fusion positioning), FasalGrow (AI remote-sensing for agriculture) - and act as the primary technical point of contact for leadership and external partners.

Research Engineer / Technical Lead

Jul 2021 - Mar 2024

National Center of Artificial Intelligence, NED University of Engineering & Technology - Karachi

- Directed applied-research programs in autonomous systems, perception, and decision-making under uncertainty; set technical direction and managed research engineers across multiple funded projects.
- Designed and built robotic platforms end to end - mechanical, electrical, and embedded-AI integration - for the Weeding Robot, Jazari Humanoid Robot, and Reverse Vending Machine.
- Implemented sensor fusion, embedded AI, and real-time control for autonomous platforms operating in low-resource, real-world environments.
- Engineered a deep-learning waste-recognition system on low-cost, lightweight embedded hardware (Reverse Vending Machine; published, Recycling / MDPI) and a computer-vision-guided weeding robot for precision agriculture (published, Machines / MDPI).

- Industrial co-supervisor for an undergraduate thesis (CV-based dairy-farm health-monitoring system), mentoring junior engineers on system design and delivery.

Research Assistant / Teaching Assistant

Aug 2019 - Jun 2021

Fab Lab & Department of Electrical Engineering, Sukkur IBA University - Sindh

- Designed and built a LoRa-based industrial SOS emergency-alarm device for hazardous environments, from circuit design through firmware and field testing.
- Designed and implemented a kinetic display system with synchronised multi-unit control (undergraduate thesis) - mechanical actuation, embedded control, and cross-unit timing.
- Contributed to additional embedded / IoT research projects including an energy-metering device and an open-source quadcopter platform.
- Teaching Assistant for Embedded Systems, mentoring students through hands-on embedded design projects.

SELECTED PROJECTS

- Adaptive Multi-Sensor Fusion for Robust Localization in GPS-Denied Environments (MS thesis) - learned context-aware INS / GNSS fusion for bounded-drift localization through GNSS outages; context-conditioned covariance network, differentiable EKF, calibrated uncertainty / integrity monitoring, and a reproducible CARLA GNSS-degradation benchmark.
- GPS-Denied Navigation Benchmark (published, IEEE / ICIICE 2025) - benchmark of nine navigation paradigms on an 8.75-hour multimodal CARLA dataset spanning weather, underground tunnels, and urban canyons; showed that architectural complexity does not guarantee better localization.
- Weeding Robot (published, Machines / MDPI) - field-deployable precision-agriculture robot combining mechanical design, embedded control, and computer-vision-based weed detection; built for low cost and easy replication.
- Reverse Vending Machine (published, Recycling / MDPI) - low-cost, lightweight embedded hardware running a deep-learning waste-classification model for a deployable recycling solution.
- Jazari Humanoid Robot - hardware, control-system, and AI-perception integration for a research-grade humanoid platform.
- Knowledge-Graph + GraphRAG Document Q&A System - self-built retrieval-augmented question answering over a Neo4j knowledge graph with LLM APIs, embeddings, and semantic memory.
- FasalGrow, Indoor Asset Tracking & Audience Insight (TPL Trakker) - production AI products for agriculture remote sensing, enterprise sensor-fusion positioning, and real-time on-device analytics.

EDUCATION

MS, Artificial Intelligence

2024 - 2026

NED University of Engineering & Technology, Karachi, Pakistan Coursework: Machine Learning & Deep Learning, Intelligent System Design, Advanced AI, Data Security & Audit, Big Data Computing.

BE, Electrical & Electronics Engineering

2015 - 2018

Sukkur IBA University, Sindh, Pakistan Relevant coursework: Control Systems, Robotics, Embedded Systems, Signal Processing.

PUBLICATIONS

- Hussain, A. & Khurram, M. (2025). Benchmarking navigation paradigms in GPS-denied environments: insights from simulated multi-modal sensor data. ICIICE, Abu Dhabi University (IEEE Xplore).
- Hussain, A., Fatima, H. S., Zia, S. M., Hasan, S., Khurram, M., Stricker, D. & Afzal, M. Z. (2023). Development of a cost-effective and easily replicable robust weeding machine - premiering precision agriculture in Pakistan. Machines, 11(4), 287. MDPI.
- Zia, H., Jawaid, M. U., Fatima, H. S., Hassan, I. U., Hussain, A., Shahzad, S. & Khurram, M. (2022). Plastic waste management through development of a low-cost and lightweight deep-learning-based reverse vending machine. Recycling, 7(5), 70. MDPI.

CERTIFICATIONS

- Self-Driving Cars Specialization - University of Toronto (2026): autonomous-vehicle stack across perception, planning, and control.
- GPU Design in SystemVerilog - SemiRise (2026): hardware-level scheduling and memory management for parallel AI workloads.
- TensorFlow Developer Professional Certificate - DeepLearning.AI (2025): CNNs, RNNs, transfer learning.
- Hands-on Data Science and AI for Healthcare - LinkedIn Learning (2025).
- Robotics Specialization (Aerial Robotics, Computational Motion Planning, Perception) - University of Pennsylvania (2022).
- Modeling & Debugging Embedded Systems; Industrial IoT Markets & Security - University of Colorado Boulder (2020).
- Principles and Applications of Digital Fabrication - Fab Foundation USA (2019).

AWARDS & SCHOLARSHIPS

- Winner, SIBACOM'19 Hardware Exhibition - recognised for original hardware product design.
- Recipient, Fab Foundation USA Scholarship (USD 4,000) - excellence in digital fabrication and product design.
- Recipient, Sindh Endowment Fund Scholarship - merit-based, full undergraduate funding.